05 - MLE, MAP, Bayes

Pattern Recognition and Machine Learning, Lab 5

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Recap

What do we know about the classes?

Bayesian task

- $p(x \mid k)$ **known**
- $p(k), W(k, d)$ **known**

We know **everything**.

Parzen windows

- $p(x \mid k)$ **unknown**
- **any** shape

Non-parametric: histogram, k-NN, kernels.

last lab

Today

- $p(x \mid k)$ **unknown**
- **known form** $p(x \mid \theta)$, unknown θ

Parametric: ML, MAP, Bayes.

today

! We estimate a few **parameters** instead of the whole density, e.g. $\theta = (\mu, \sigma^2)$ of a normal distribution.

The setting

Known form, unknown parameters:

$$p(x \mid k) = p(x \mid k, \theta_k)$$

Each class separately, as with Parzen: we drop k and write $p(x \mid \theta)$.

Training set, samples **i.i.d.**:

$$T = \{x_1, x_2, \ldots, x_N\}$$

Likelihood of the parameters:

$$L(\theta) = p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$$

Log-likelihood, same maximum:

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{N} \ln p(x_i \mid \theta)$$

> **!** $L(\theta)$ is a function of θ for **fixed data**, **not** a distribution over θ.

Likelihood: try it

μ 1.2

σ 1.6

☑ density ☑ log-likelihood

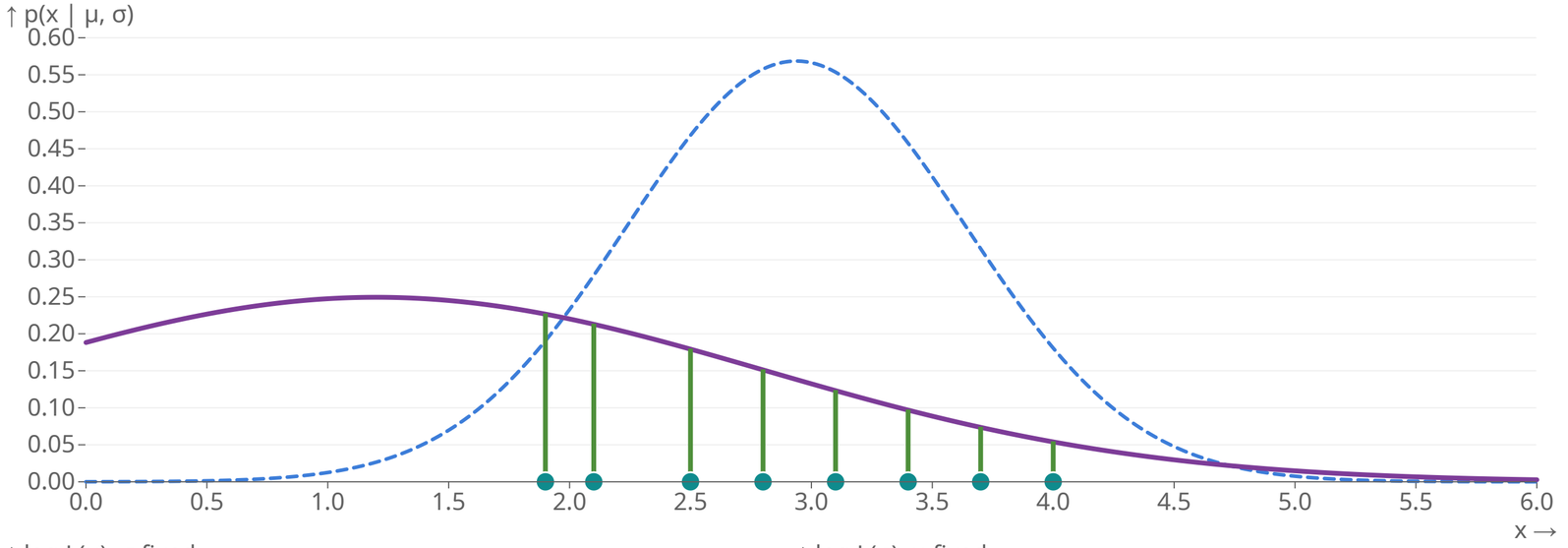
☑ ML solution

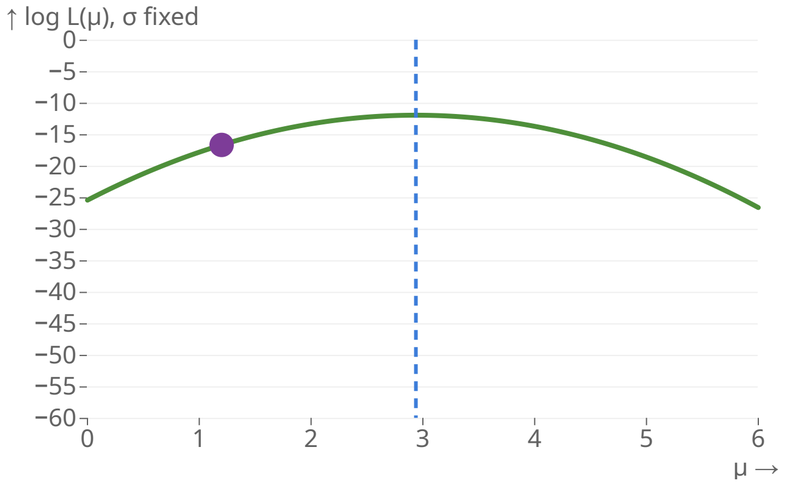
N = 8 samples

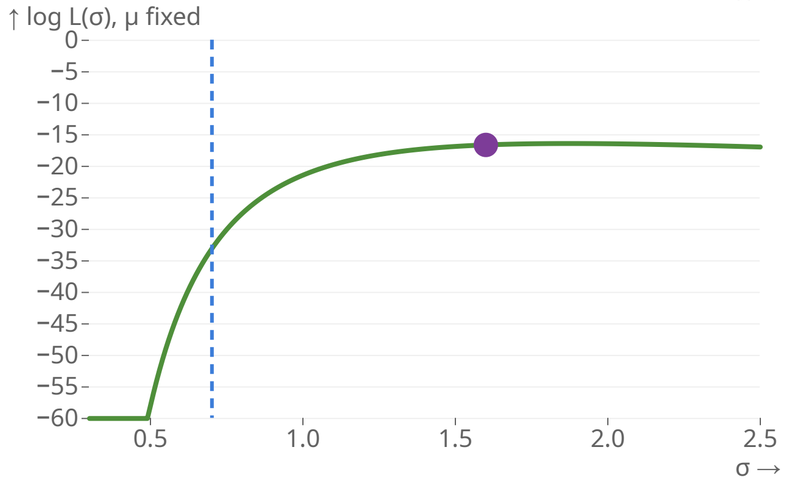
log L(μ, σ) = **-16.60**

ML: μ = 2.94, σ = 0.70

log L = -8.52







Three ways to estimate θ

Maximum likelihood (ML)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

Choose the parameters under which the data are **most probable**:

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} p(T \mid \theta) = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i \mid \theta)$$

Prediction for a new x^*: **plug in** the estimate, $p(x^* \mid T) \approx p(x^* \mid \hat{\theta}_{\mathrm{ML}})$

Result: $\hat{\theta}_{\mathrm{ML}}$, **one number** (a point estimate), not a distribution.

Pros

Simple, no extra assumptions, the best with **a lot of data**.

Cons

With **few** data it trusts them blindly: 3 tosses, 3 heads, so $\hat{\pi}_{\mathrm{head}} = 1$?

Maximum a posteriori (MAP)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

θ is a **random variable** with a **prior** $p(\theta)$. Maximize the **posterior**:

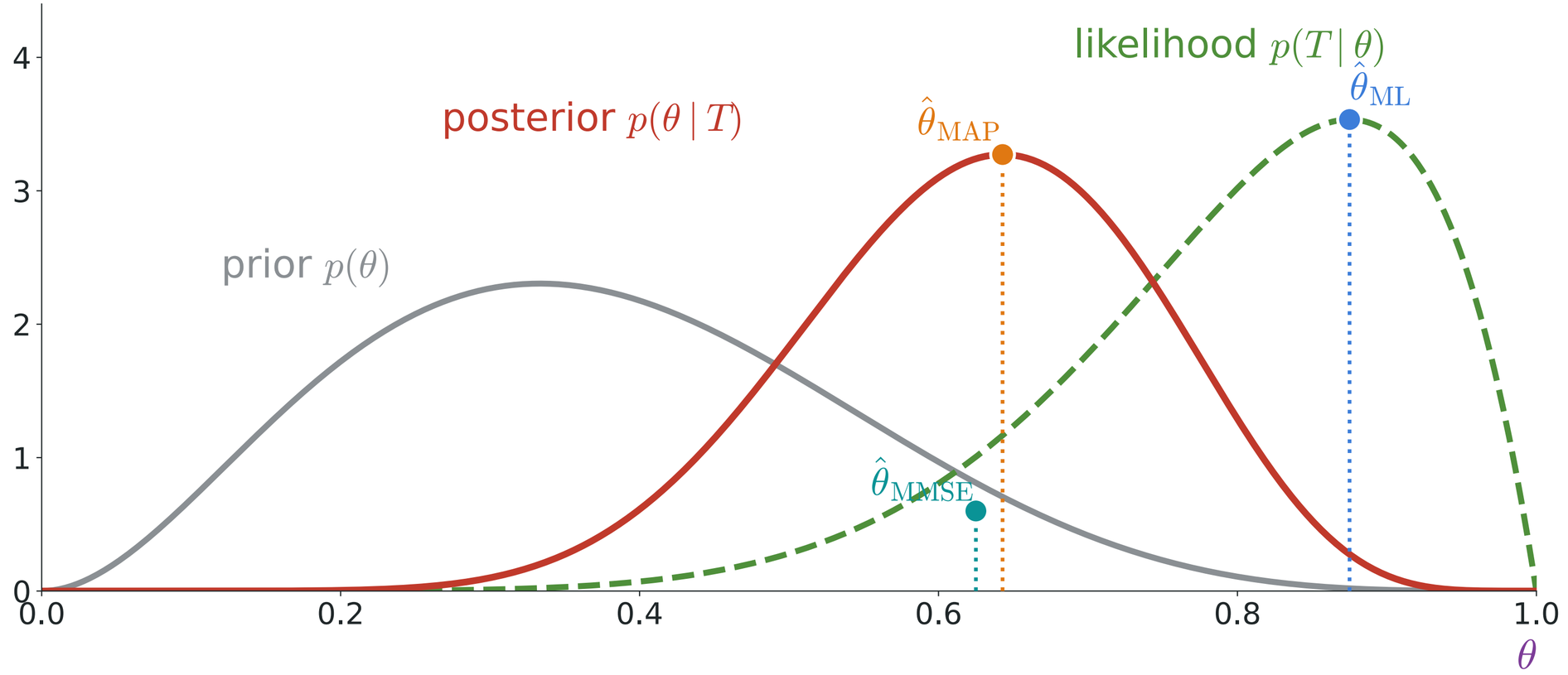
$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid T) = \arg\max_{\theta} \frac{p(T \mid \theta)\, p(\theta)}{p(T)} = \arg\max_{\theta} p(T \mid \theta)\, p(\theta)$$

- $p(T)$ does not depend on θ: drop it
- the prior $p(\theta)$ has to be **specified**

Result: $\hat{\theta}_{\text{MAP}}$, again **one number**.

Convenient prior: **conjugate**. The posterior has the same form as the prior.

Prior, likelihood, posterior



Posterior $\propto$ likelihood $\times$ prior: a compromise between the two.

Bayesian inference: keep the whole posterior

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

Do not pick one θ, keep the **whole distribution**:

$$p(\theta \mid T) = \frac{p(T \mid \theta)\, p(\theta)}{p(T)}$$

Prediction: **average** all models, weighted by their probability:

$$p(x^* \mid T) = \int p(x^* \mid \theta)\, p(\theta \mid T)\, d\theta$$

Result: **a distribution** $p(\theta \mid T)$, no single θ. The prediction is a distribution too.

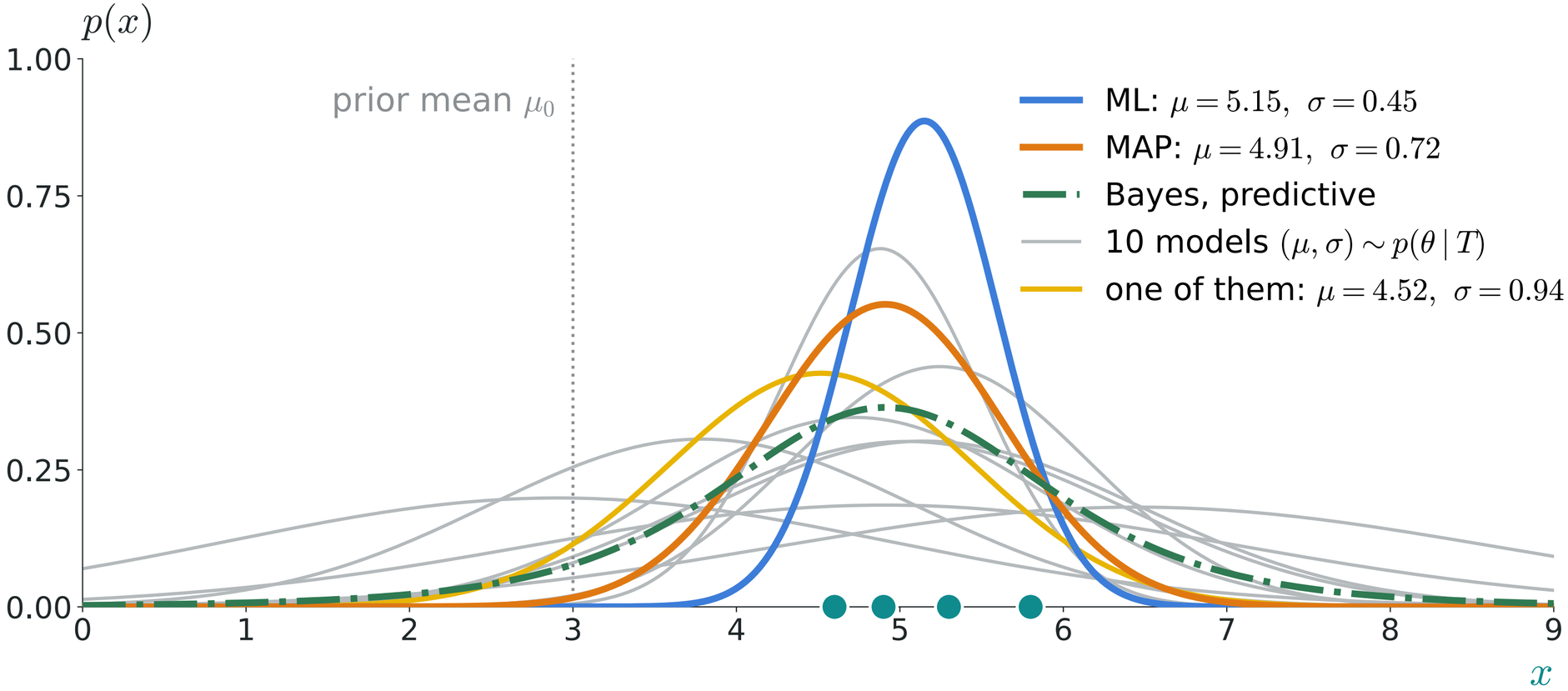
Pros

Uses all the information, knows how **uncertain** it is.

Cons

The integral: in closed form only for **conjugate** priors.

Comparing the estimates



It averages models from $p(\theta \mid T)$: 10 of them drawn, **one** highlighted.

Bayesian parameter estimation (MMSE)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

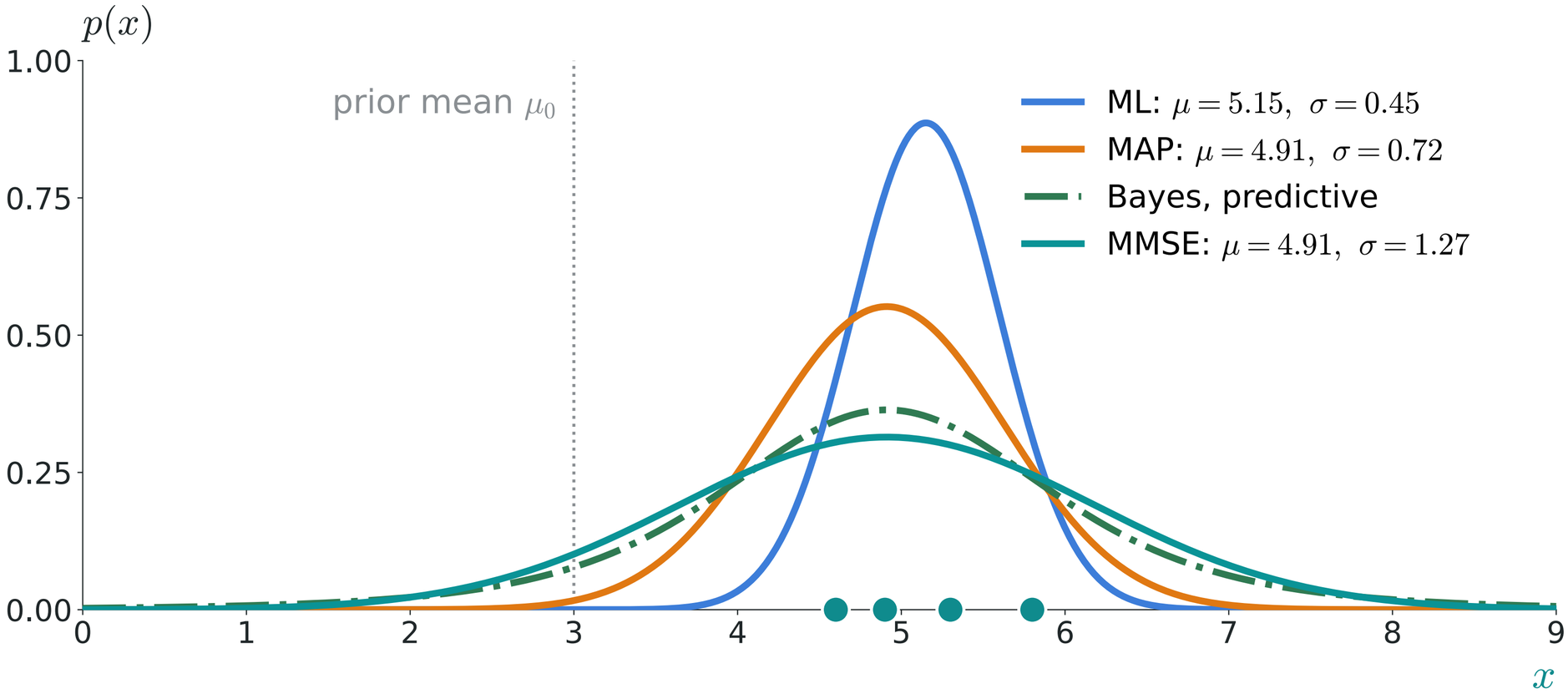
Or: pick **one** $\hat{\theta}$ with the smallest expected **squared error**:

$$R(\hat{\theta}) = \int p(\theta \mid T)\,(\theta - \hat{\theta})^2\, d\theta$$

$$\frac{\partial R}{\partial \hat{\theta}} = -2 \int p(\theta \mid T)\,(\theta - \hat{\theta})\, d\theta = 0 \quad \Rightarrow \quad \hat{\theta}_{\mathrm{MMSE}} = \int \theta\, p(\theta \mid T)\, d\theta$$

Result: $\hat{\theta}_{\mathrm{MMSE}} = \mathbb{E}[\theta \mid T]$, **one number**: the mean of the posterior. Prediction by plugging in.

Comparing the estimates: MMSE



MMSE: plug in the posterior mean. Still a normal, unlike **Bayes**.

Three estimates, one picture

	prior	result	prediction $p(x^* \mid T)$
ML	no	$\hat{\theta}_{\mathrm{ML}}$, max of likelihood	$p(x^* \mid \hat{\theta}_{\mathrm{ML}})$
MAP	yes	$\hat{\theta}_{\mathrm{MAP}}$, max of posterior	$p(x^* \mid \hat{\theta}_{\mathrm{MAP}})$
MMSE	yes	$\hat{\theta}_{\mathrm{MMSE}}$, mean of posterior	$p(x^* \mid \hat{\theta}_{\mathrm{MMSE}})$
Bayes	yes	the whole $p(\theta \mid T)$	$\int p(x^* \mid \theta)\, p(\theta \mid T)\, d\theta$

With a **flat** prior, MAP equals ML.

With **a lot of data**, all four agree.

Conjugate priors

likelihood $p(x \mid \theta)$	parameter	conjugate prior	posterior
binomial	π	Beta (a, b)	Beta $(a + R,\ b + N - R)$
categorical	$p_1, \ldots, p_K$	Dirichlet $(\alpha_1, \ldots, \alpha_K)$	Dirichlet $(\alpha_k + N_k)$
normal	μ (σ^2 known)	normal (μ_0, σ_0^2)	normal
normal	μ, σ^2	NIG	NIG assignment
Poisson	λ	Gamma (α, β)	Gamma $(\alpha + \sum x_i,\ \beta + N)$
uniform $[0, \theta]$	θ	Pareto (θ_0, k)	Pareto $(\max(\theta_0, m),\ k + N)$

The full table: Wikipedia, Conjugate prior.

The German tank problem

Counting tanks from serial numbers



WW2: the Allies read **serial numbers** on captured tanks (gearboxes, wheels). How many tanks does Germany produce per month?

month	intelligence	statistics	German records
June 1940	1000	169	**122**
June 1941	1550	244	**271**
August 1942	1550	327	**342**

Ruggles and Brodie (1947); table from Wikipedia, *German tank problem*.

? We have seen n tanks, the largest serial number is m. How many tanks are there?

The model: uniform distribution

Serial numbers are uniform on $[0, \theta]$, θ is the **number of tanks**:

$$p(x \mid \theta) = \begin{cases} \frac{1}{\theta} & 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}$$

Continuous for simplicity, real serial numbers are integers.

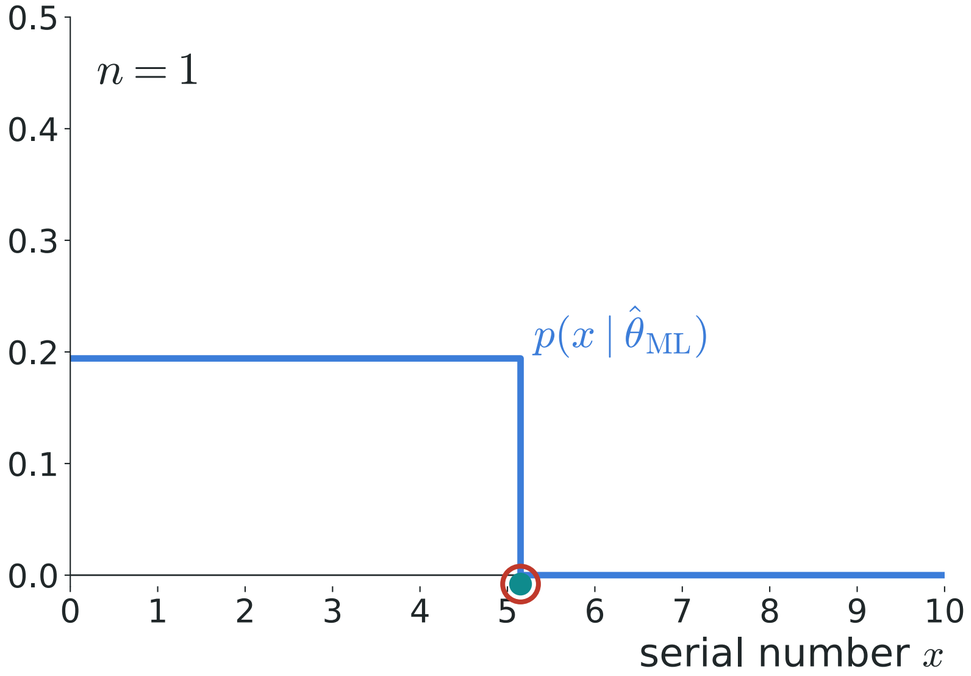
$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} [\![x_i \le \theta]\!]$$

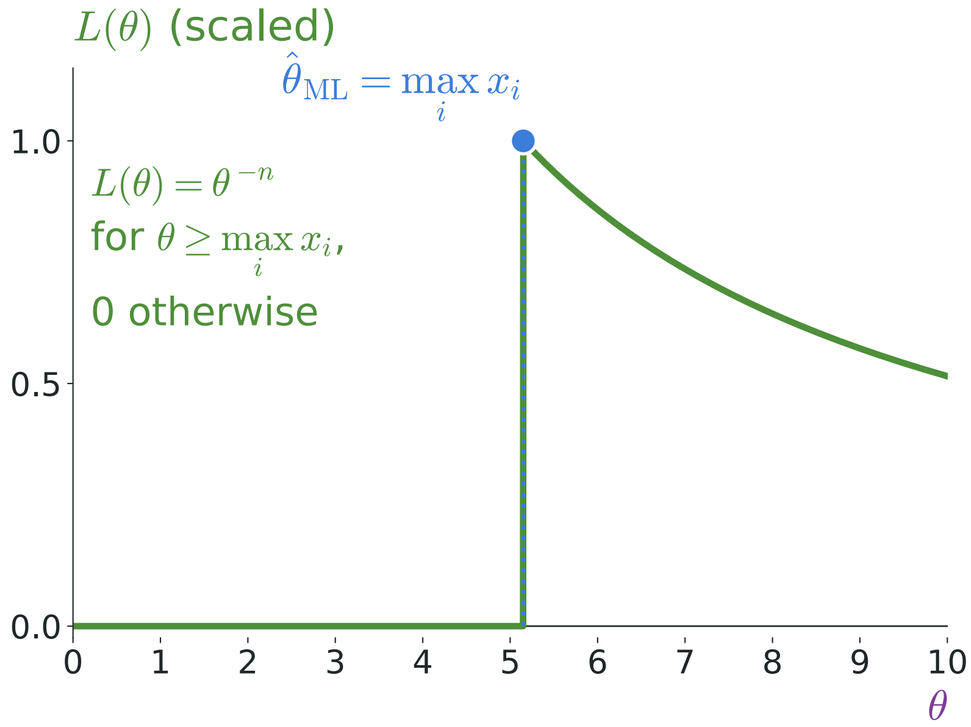
$$L(\theta) = \begin{cases} \theta^{-n} & \theta \ge \max_i x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{\theta}_{\text{ML}} = \max_i x_i = m$$

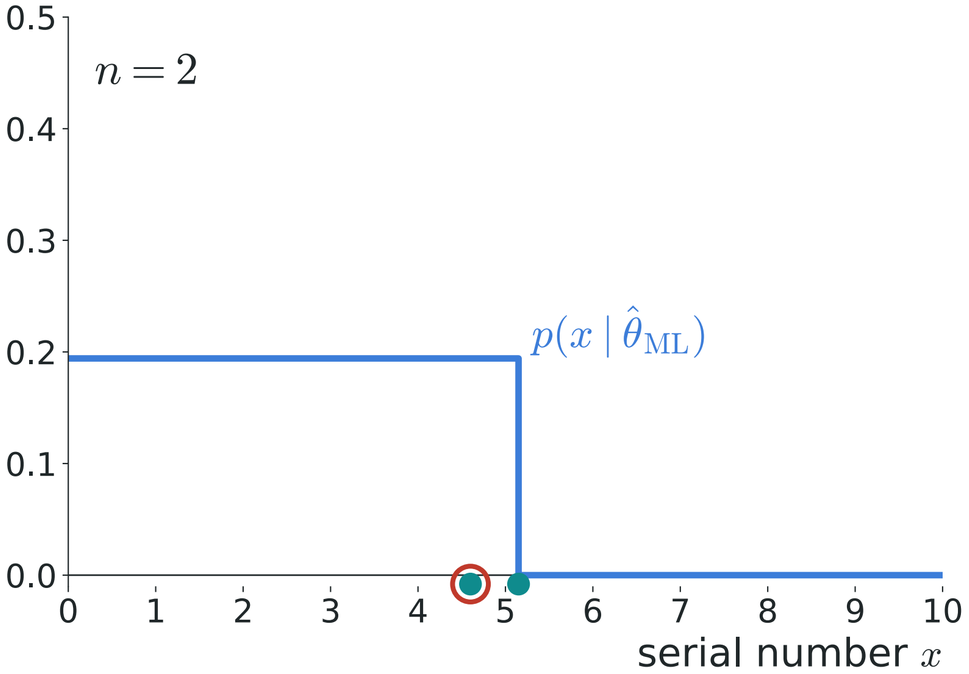
? Is it a good estimate? Can the true θ be **smaller** than m?

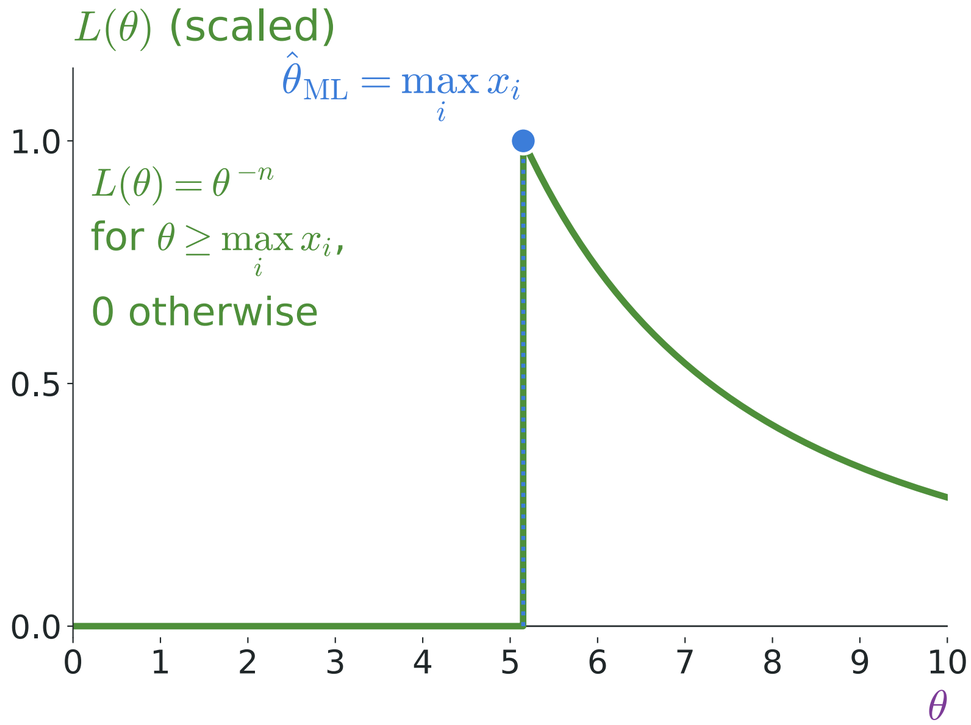
ML with more and more tanks



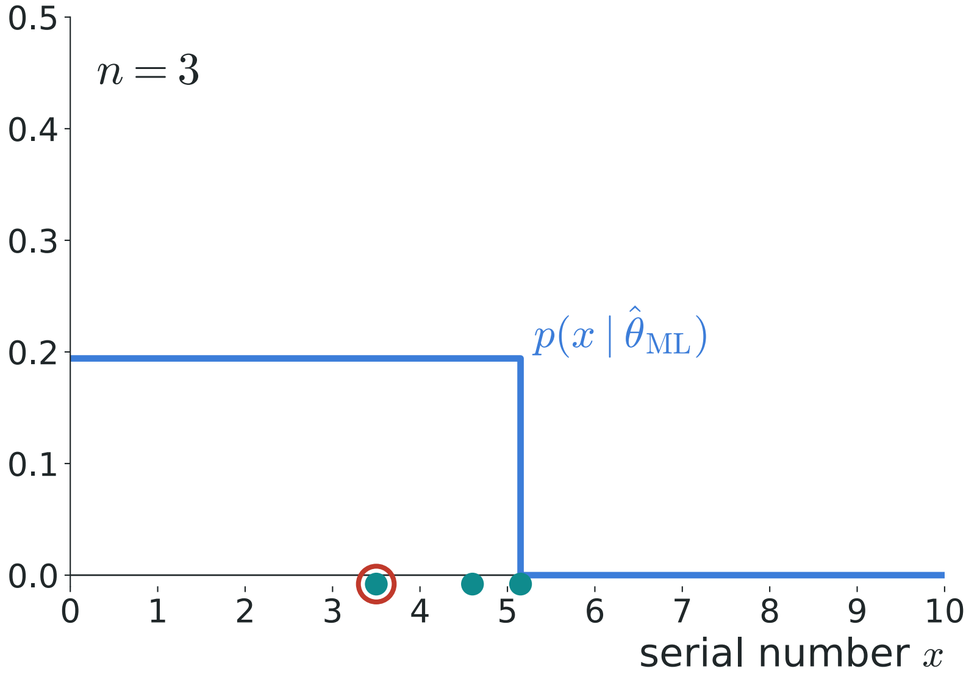


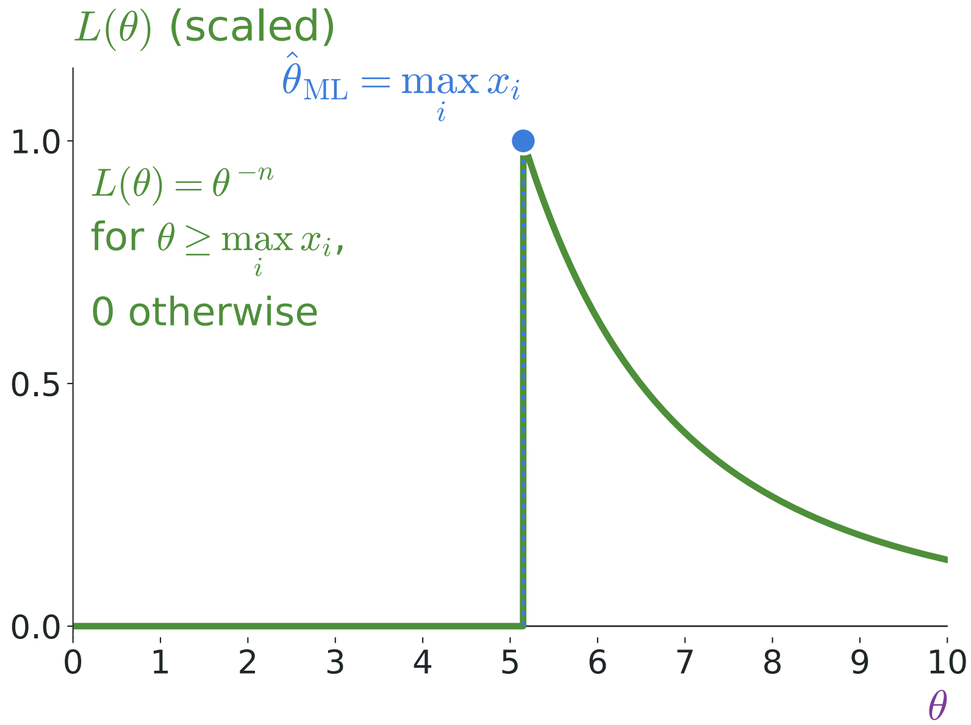
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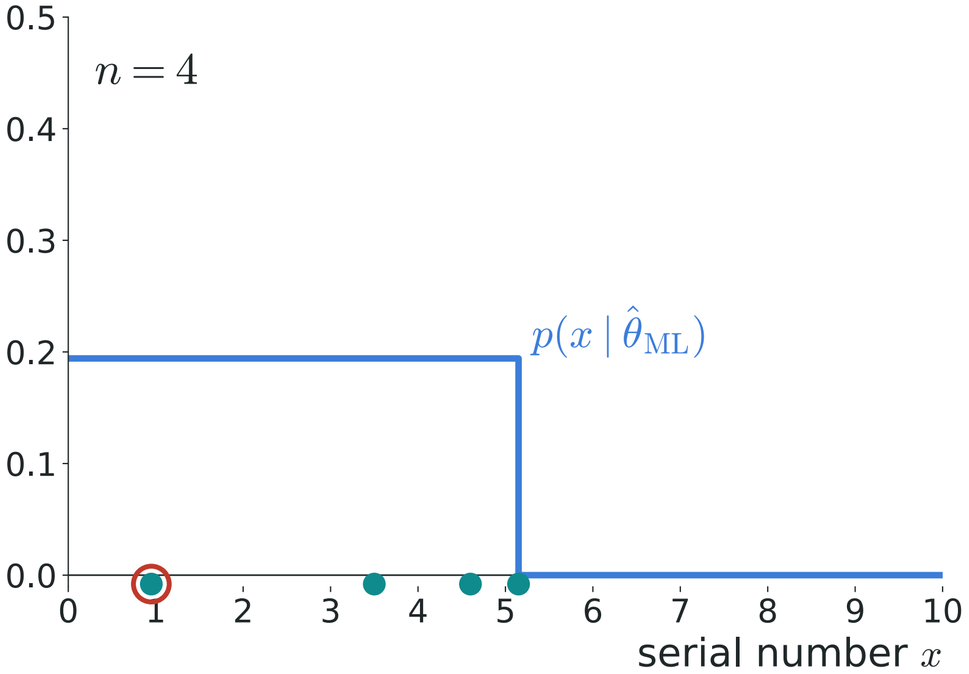


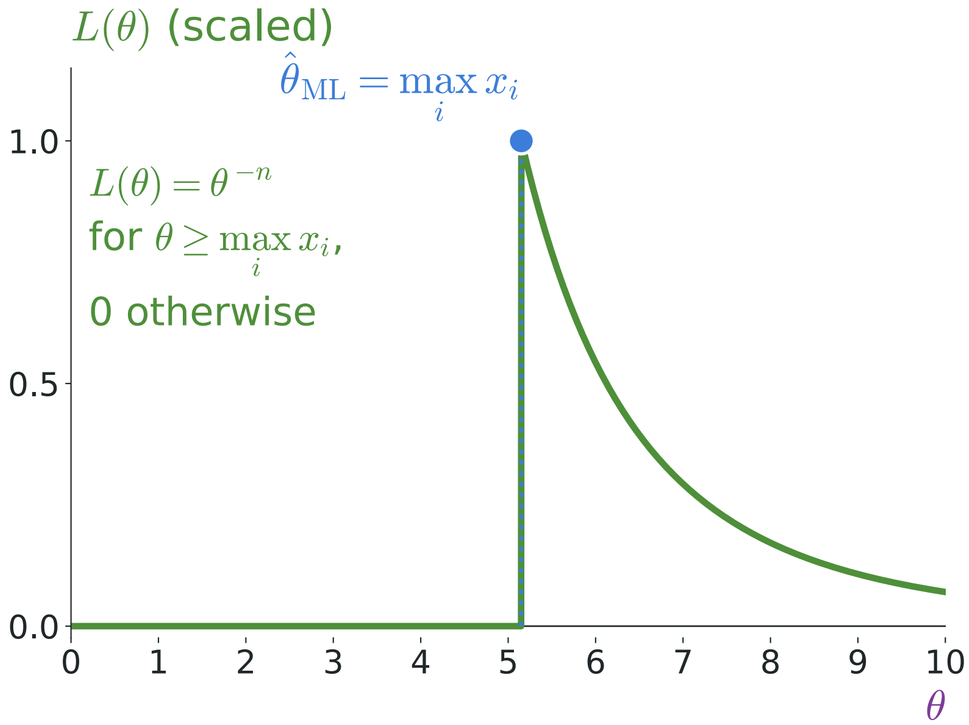
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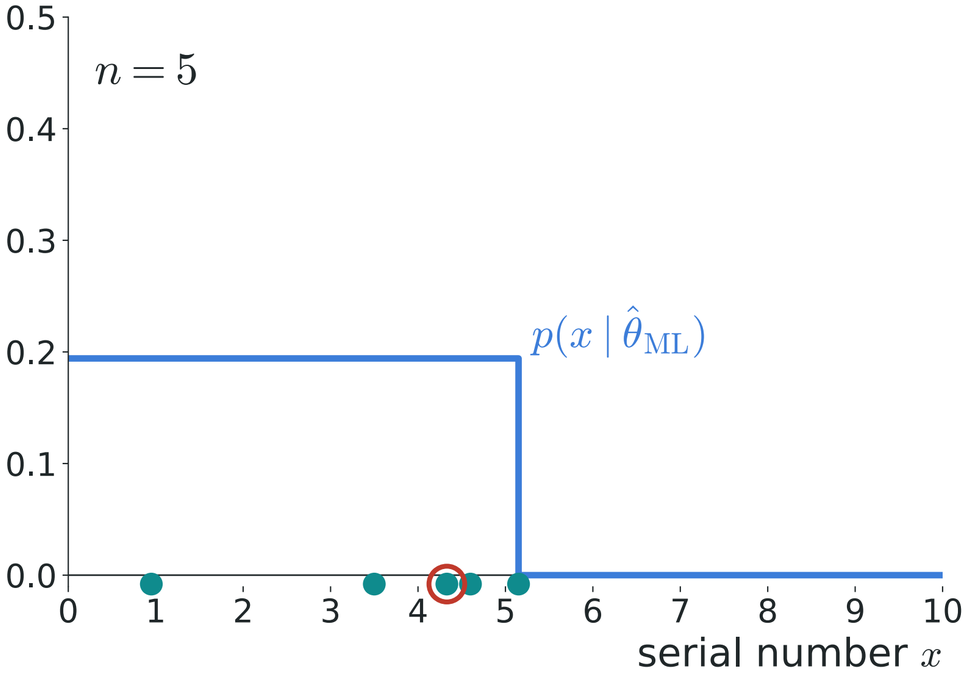


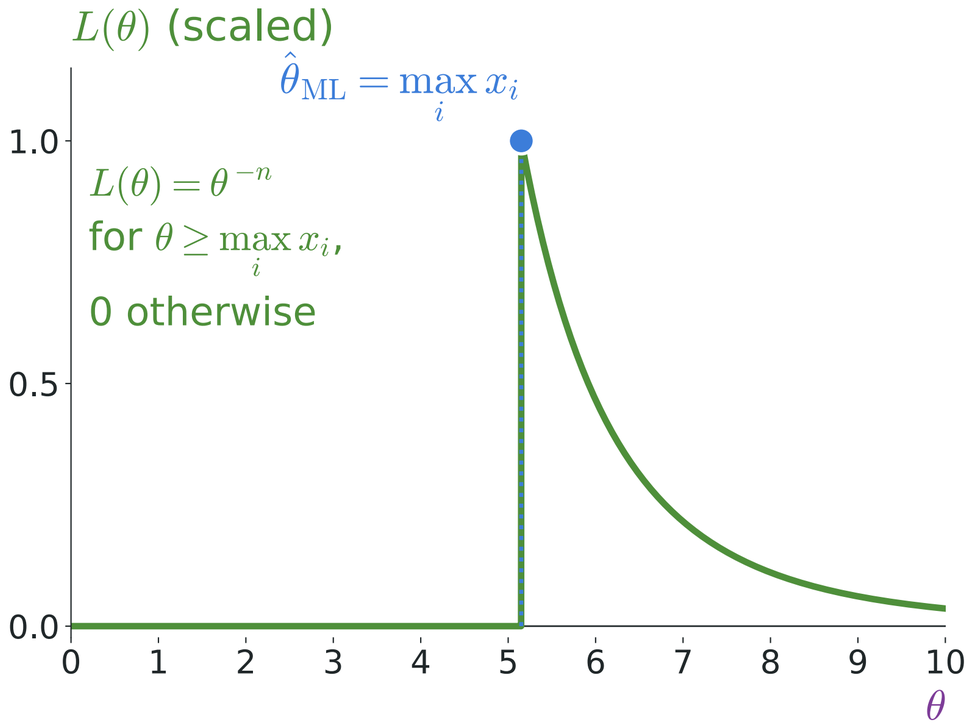
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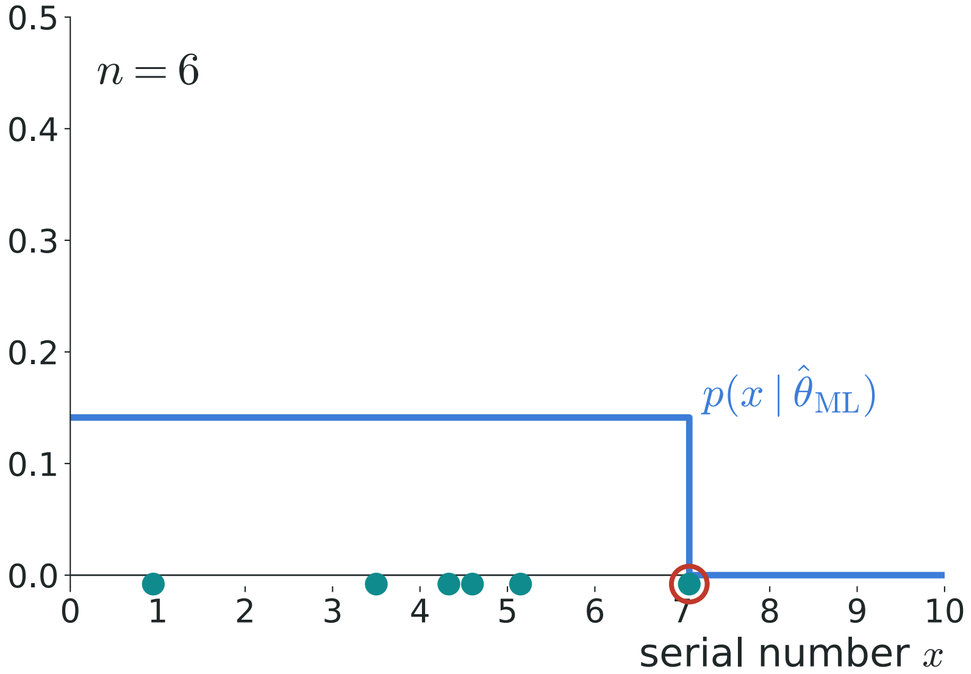


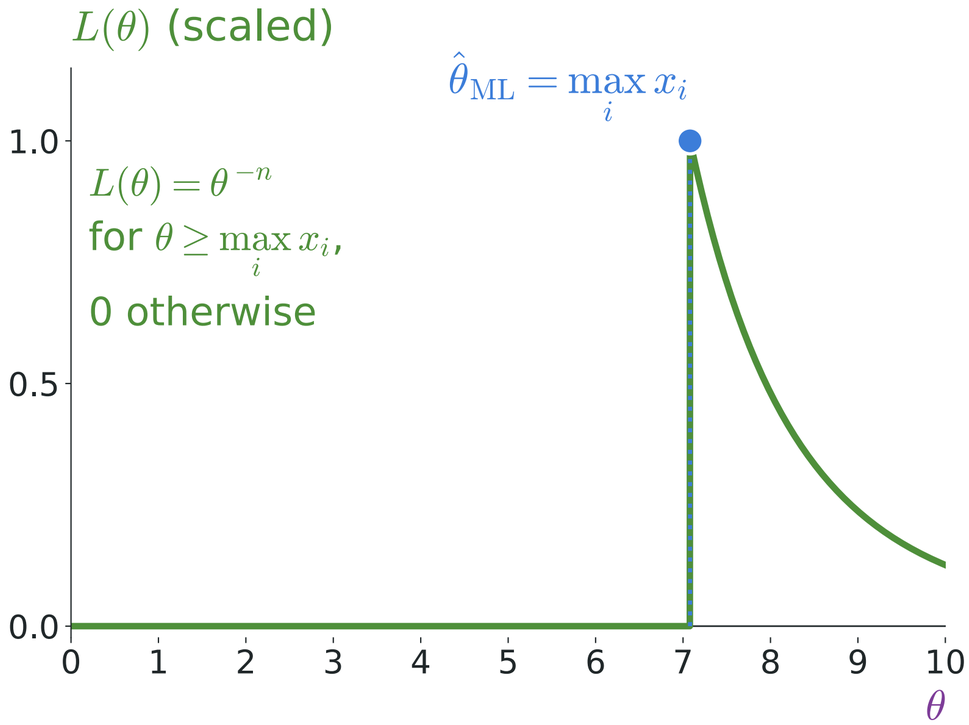
ML with more and more tanks



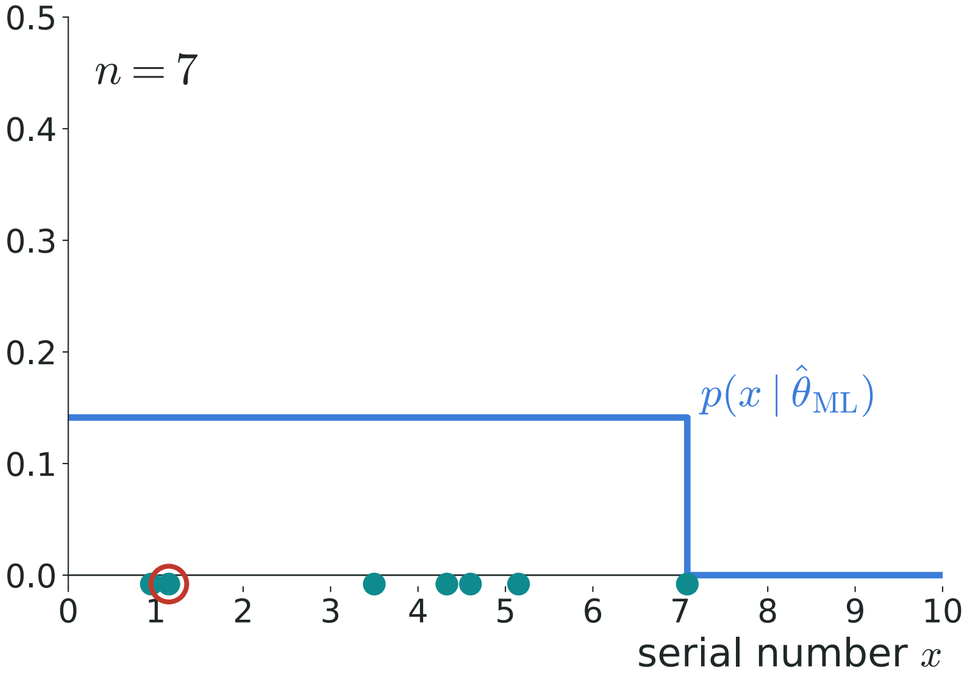


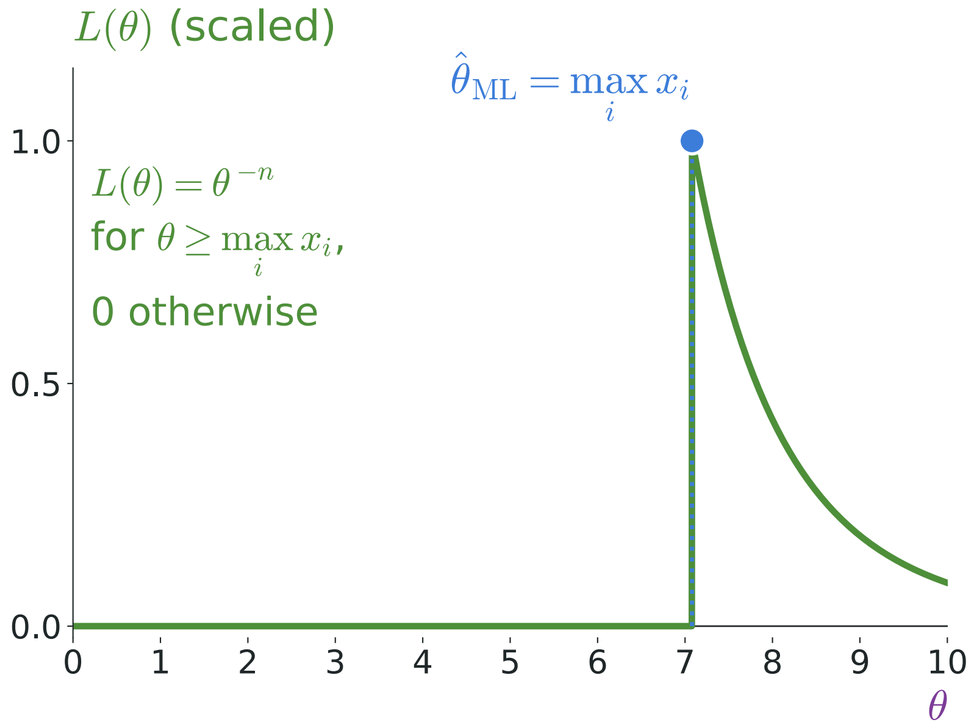
ML with more and more tanks



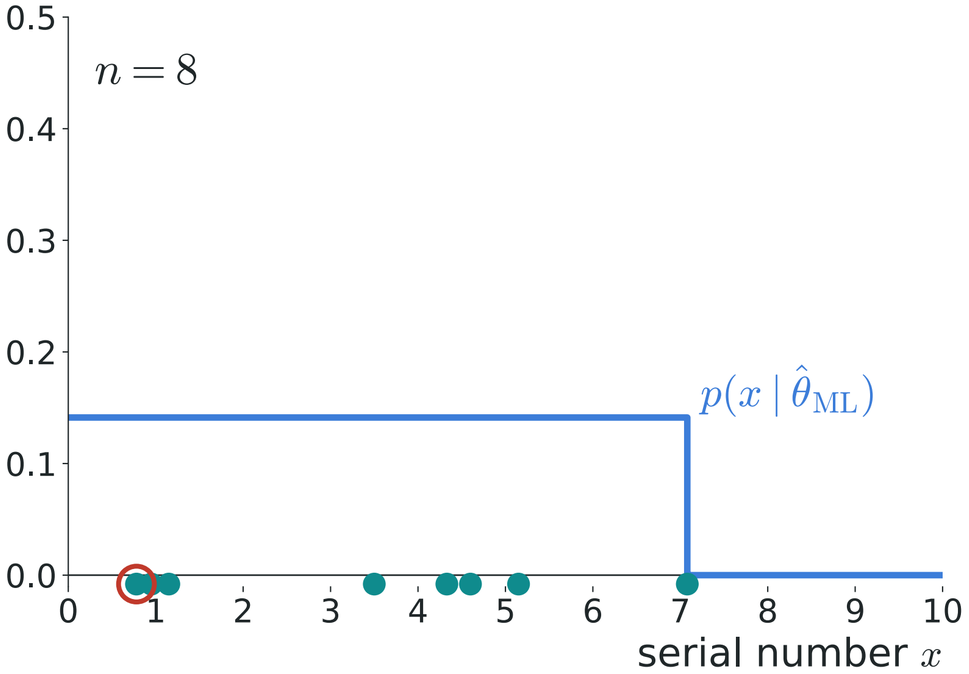


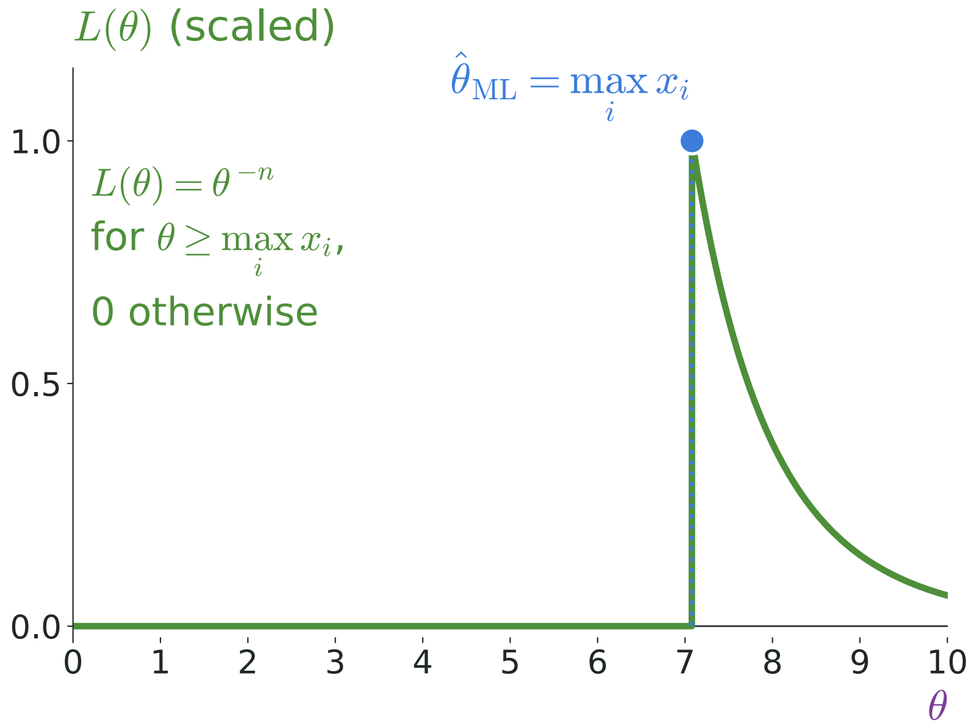
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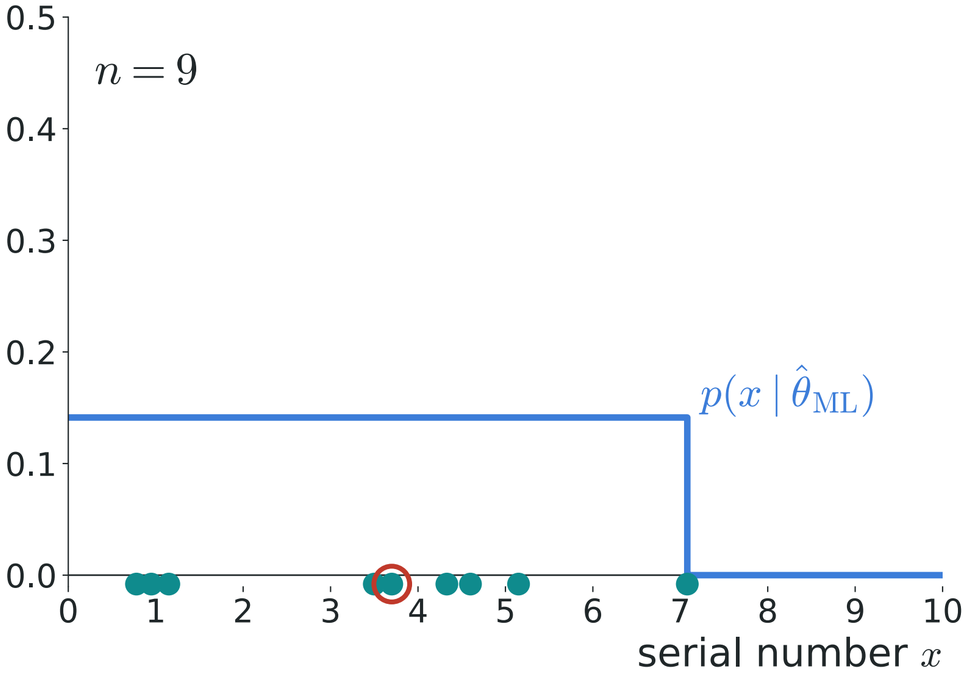


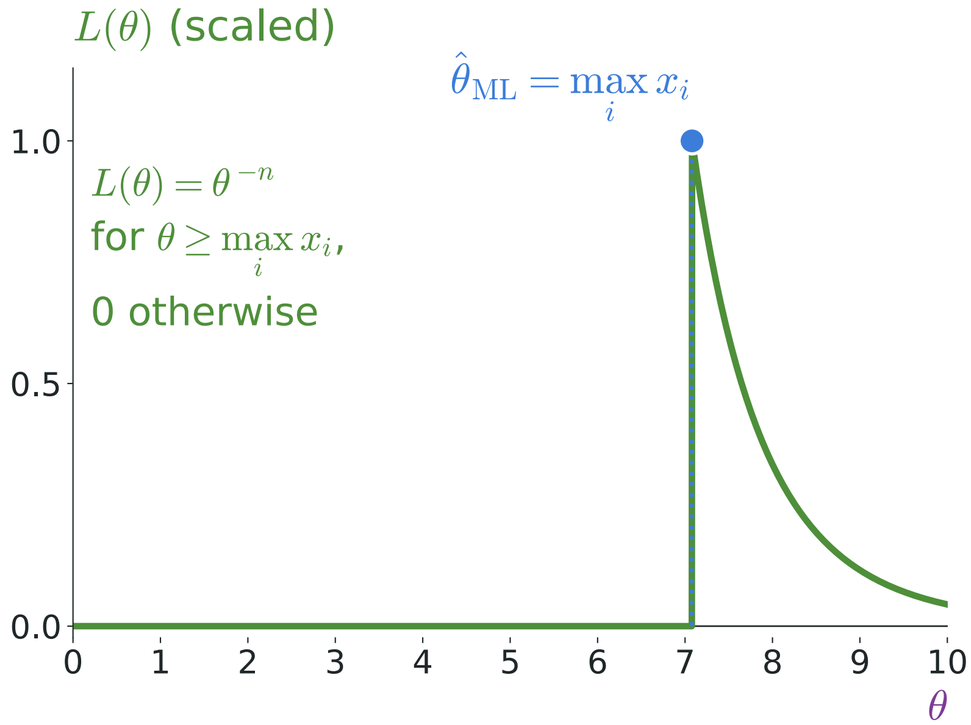
ML with more and more tanks



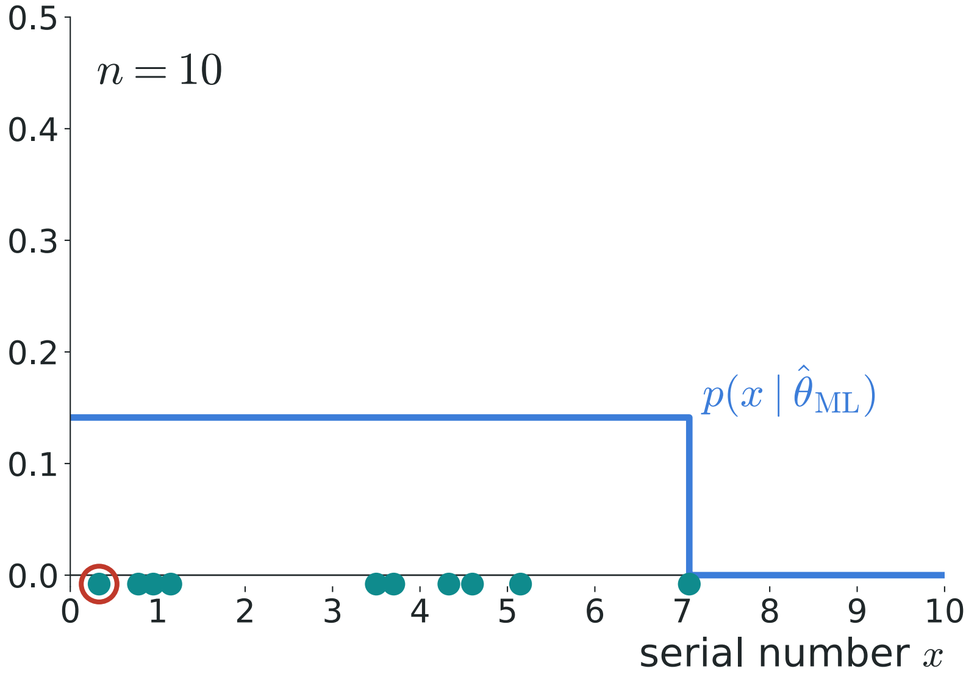


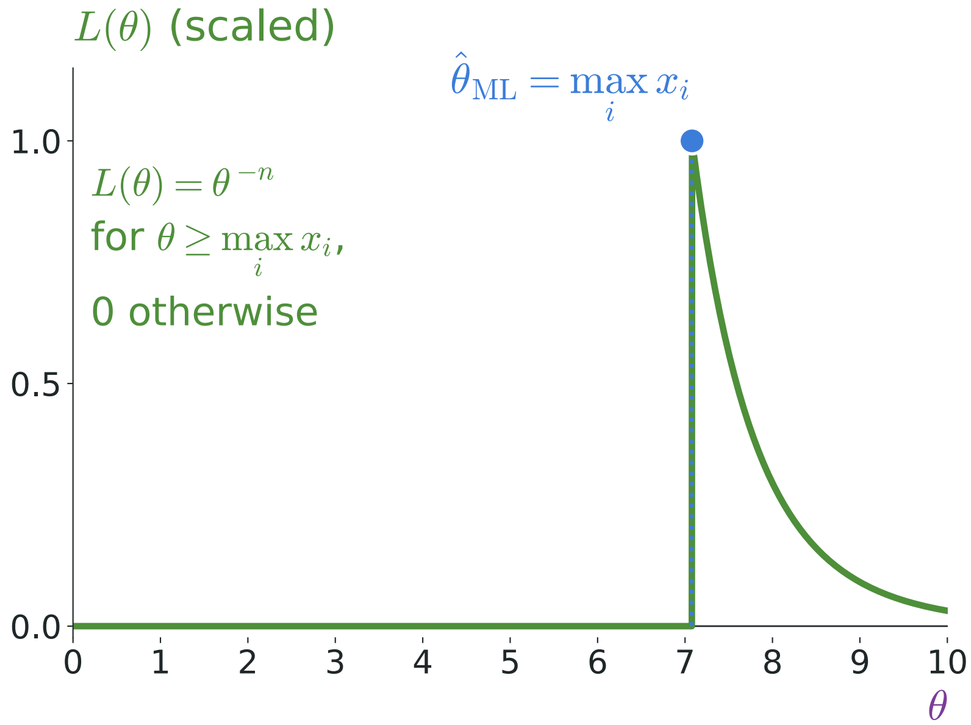
ML with more and more tanks





ML with more and more tanks





Derivation: Bayes for the tanks

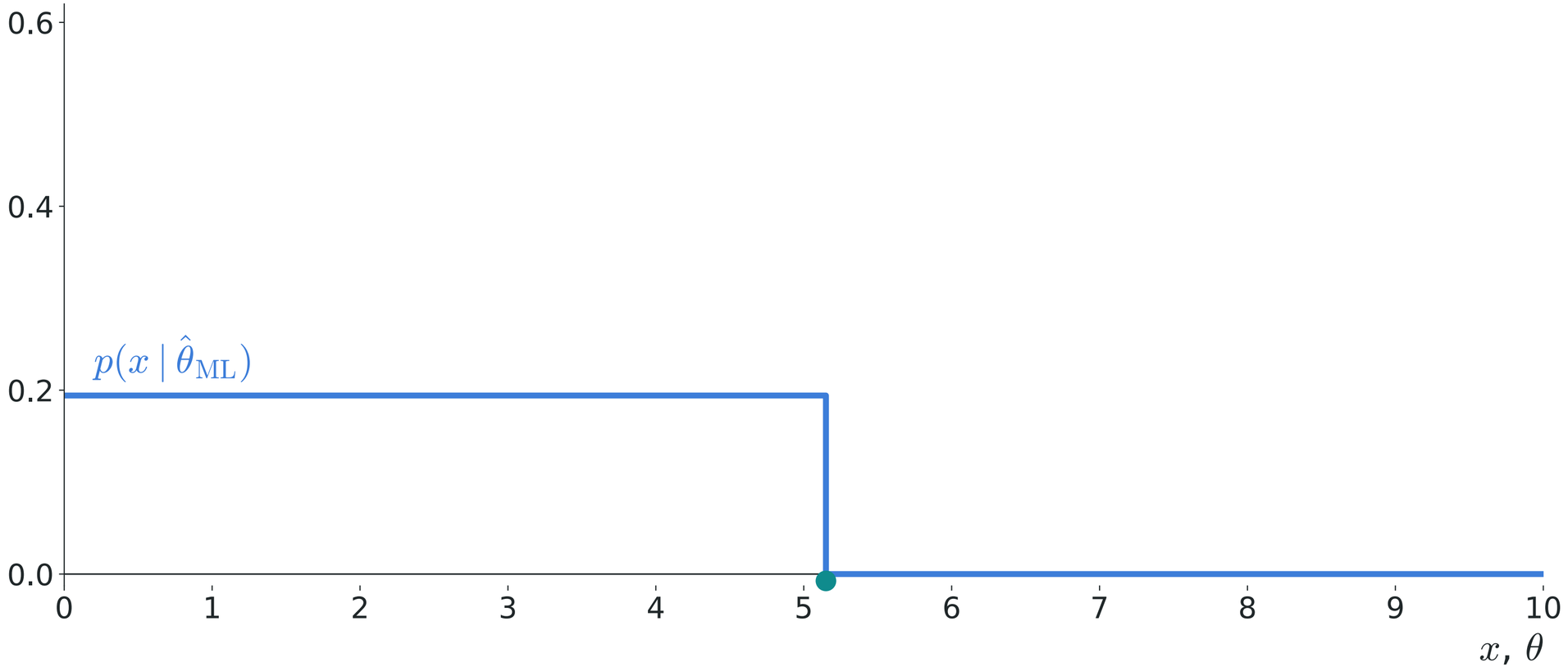
Prior: θ uniform on $[0, M]$, at most M tanks. Posterior, $m = \max_i x_i$:

$$p(\theta \mid T) \propto \prod_{i=1}^{n} \frac{1}{\theta} [\![x_i \le \theta]\!] \, \frac{1}{M} [\![\theta \le M]\!] \quad \Rightarrow \quad p(\theta \mid T) = \frac{1}{Z} \frac{1}{\theta^n} [\![m \le \theta \le M]\!]$$

$$Z = \int_m^M \frac{d\theta}{\theta^n} = \frac{1}{n-1} \left(\frac{1}{m^{n-1}} - \frac{1}{M^{n-1}} \right), \qquad \hat{\theta}_{\mathrm{MAP}} = m = \hat{\theta}_{\mathrm{ML}} \ \ \text{(flat prior)}$$

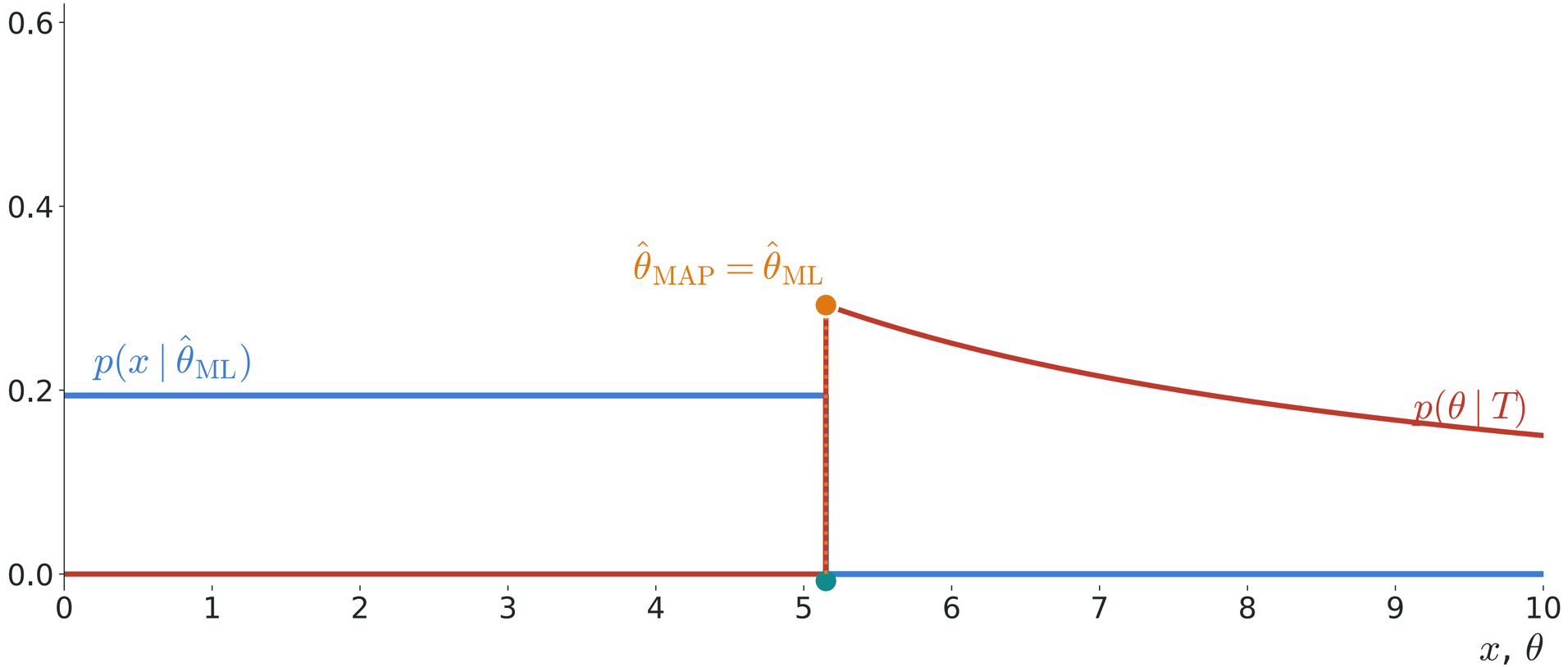
$$\hat{\theta}_{\mathrm{MMSE}} = \frac{1}{Z} \int_m^M \frac{\theta}{\theta^n} \, d\theta \ \xrightarrow{M \to \infty} \ m \, \frac{n-1}{n-2}, \qquad p(x \mid T) = \int_{\max(x,m)}^{M} \frac{1}{\theta} \, p(\theta \mid T) \, d\theta$$

ML, MAP, MMSE and Bayes for the tanks



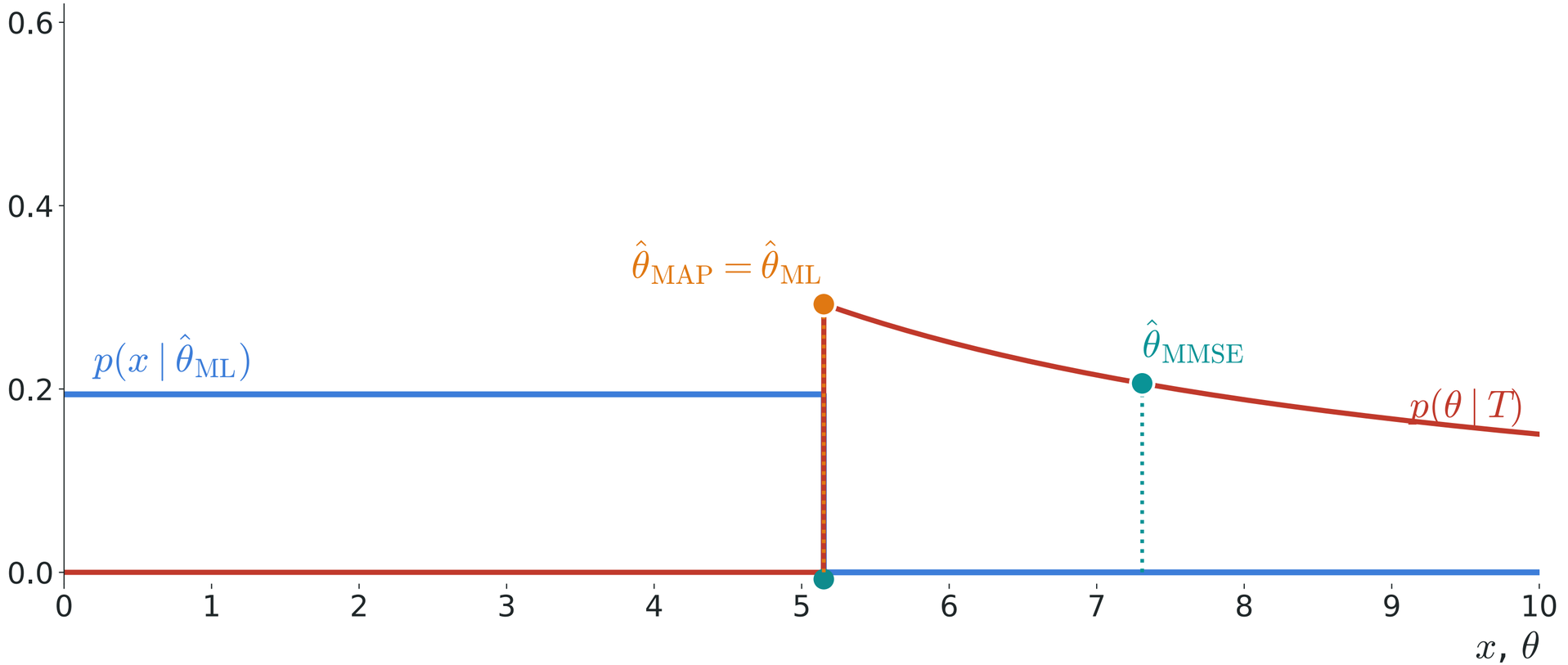
One tank, $x_1 = 5.15$: the ML plug-in density ends at the observation.

ML, MAP, MMSE and Bayes for the tanks



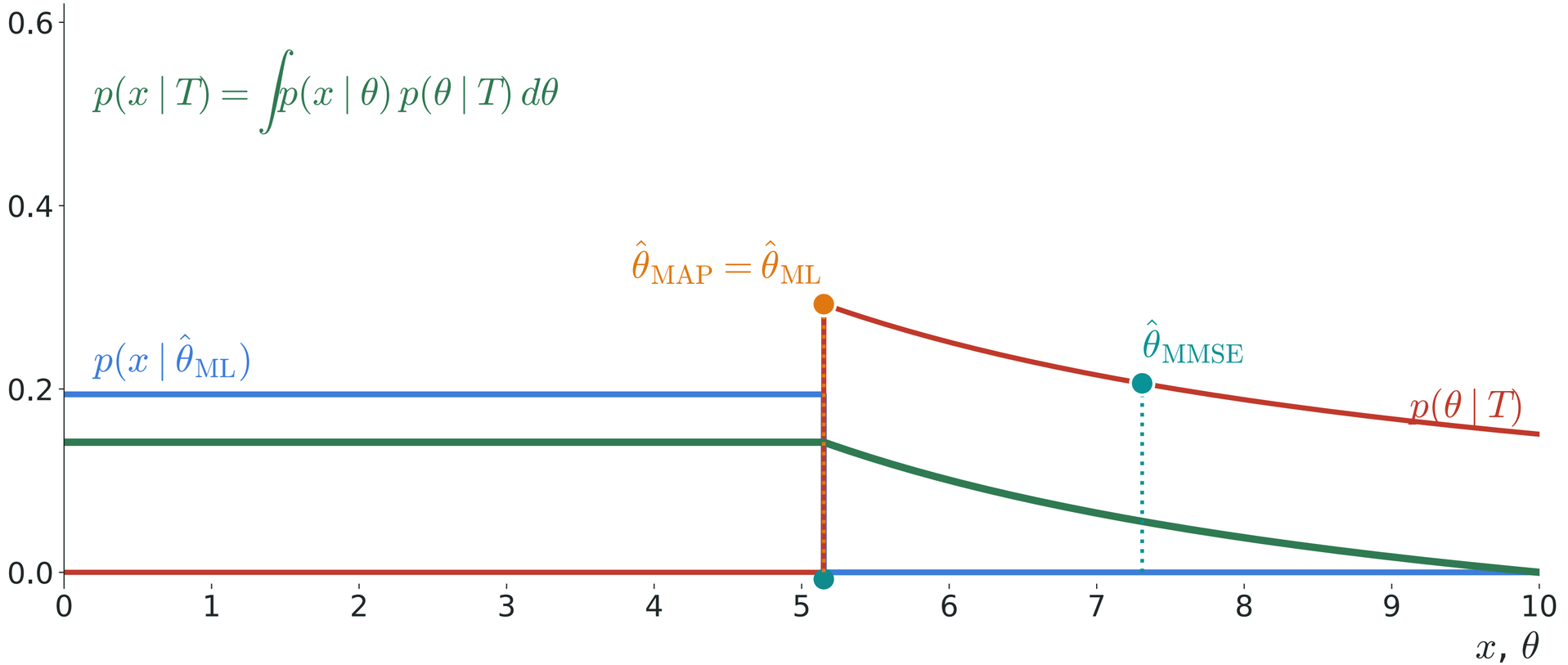
Posterior over θ (prior uniform on $[0, 10]$): its maximum is again m.

ML, MAP, MMSE and Bayes for the tanks



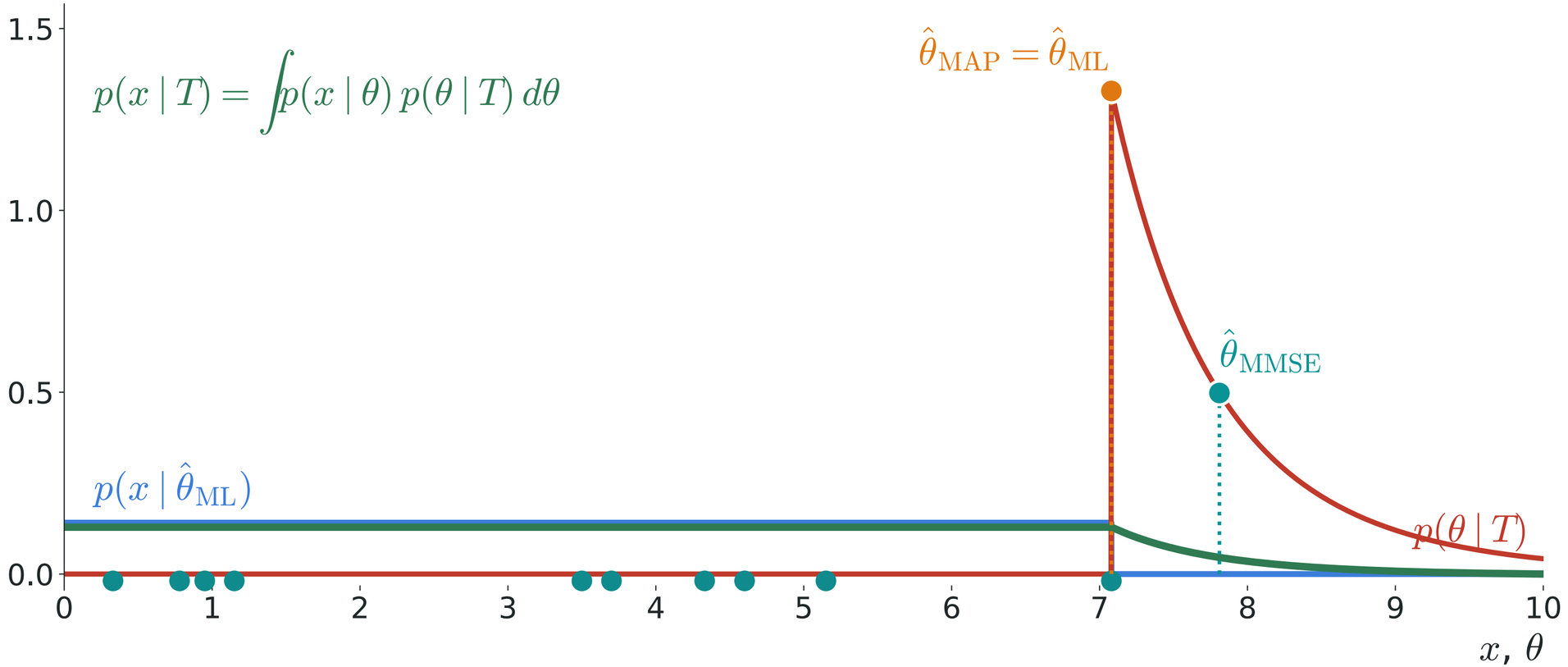
Its mean $\hat{\theta}_{\mathrm{MMSE}} = 7.3$ lies to the right of m.

ML, MAP, MMSE and Bayes for the tanks



Bayesian prediction $p(x \mid T)$: larger numbers are still possible.

ML, MAP, MMSE and Bayes for the tanks



Ten tanks: the posterior is narrow, all estimates are close.

German tanks: try it

n tanks 1

true θ 7.3

prior M 10

show
- [x] ML plug-in
- [ ] posterior
- [ ] MMSE
- [ ] predictive

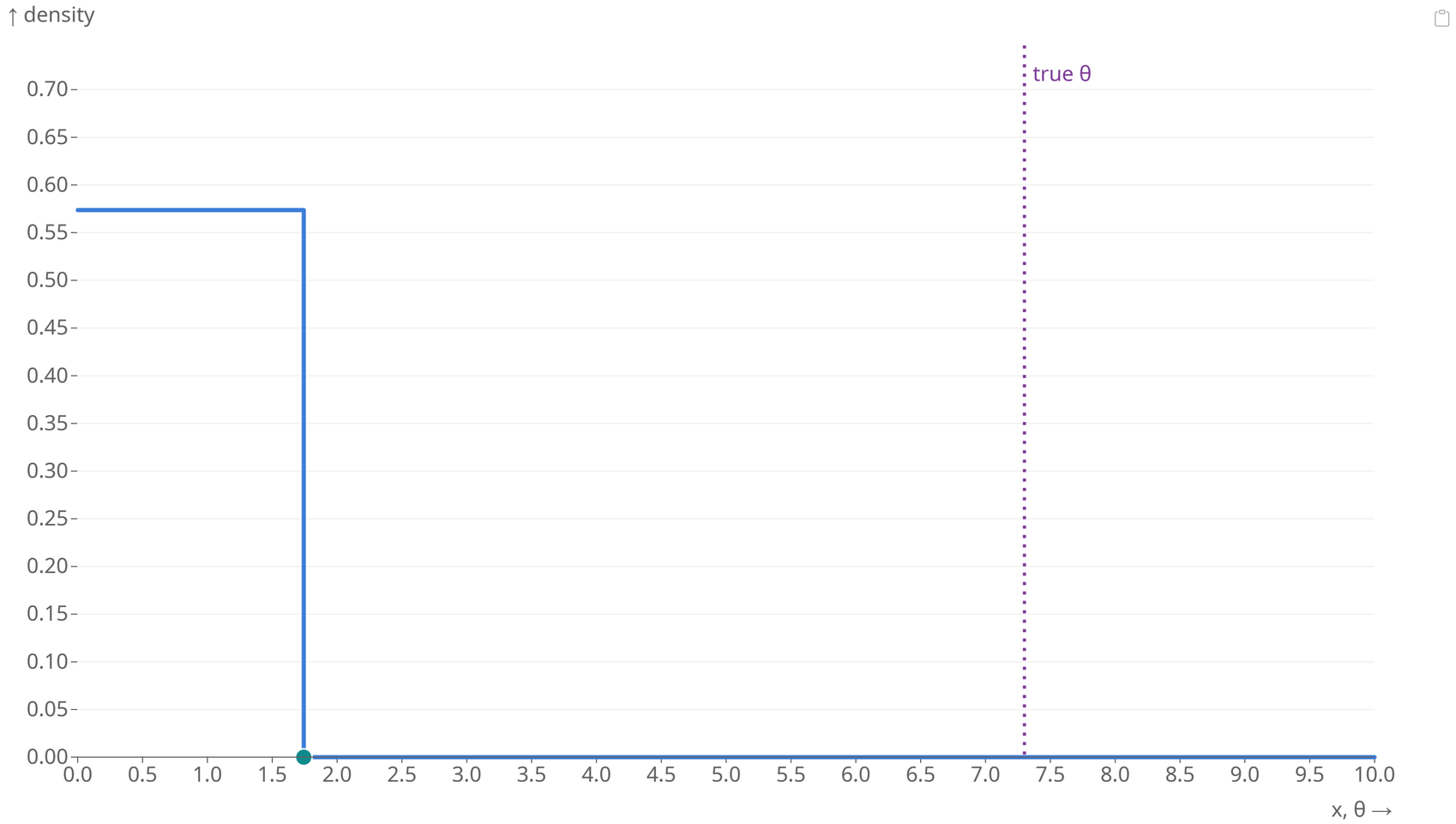
new tanks

$\theta_{ML} = \theta_{MAP} = 1.74$

$\theta_{MMSE} = 4.73$

unbiased m(n+1)/n = 3.49

P(next x > m) = 0.528



The real (discrete) problem

Serial numbers $1, 2, \ldots, N$, we see k of them, the largest is m:

estimate	formula	$k = 4, m = 60$
ML	m	60
unbiased	$m + \frac{m}{k} - 1$	74
Bayes, posterior mean	$\frac{(m-1)(k-1)}{k-2}$	89

ML is **biased**: it always underestimates. Adding the average gap $\frac{m}{k}$ fixes it.

Why is **ML** the most common?

Why ML?

Under general conditions, as $N \to \infty$

- **consistent**: converges to the true θ
- asymptotically **unbiased** and **efficient** (the smallest variance)
- asymptotically **normal**

In practice

- no prior to choose
- a lot of data: MAP and Bayes give the **same** result
- Bayes needs integrals: closed form only for conjugate priors

But with few data

- overconfident: tanks (and socks, unseen categories later)
- biased: $\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{N}\sum(x_i - \hat{\mu})^2$, $\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{N-1}{N}\sigma^2$

A bit of history

- **1769** Lagrange: ML for the multinomial gives relative frequencies
- **1778** D. Bernoulli: multiply the densities of the observations, maximize
- **1809** Gauss: posterior mode with a flat prior, gives **least squares**
- **1912, 1922** Fisher: the method gets its name and its theory

> "Who would rise to argue in favor of a method of minimum likelihood, or even mediocre likelihood?"

S. M. Stigler: *The Epic Story of Maximum Likelihood*, Statistical Science, 2007.

Example: socks

Socks

A drawer with **red** and **green** socks. We pull out a sock, note its colour and **put it back**. After N draws, R socks were red.

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{\text{red}, \text{green}\}$$

$$p(\text{red} \mid \pi) = \pi, \qquad p(\text{green} \mid \pi) = 1 - \pi$$

π: the ratio of red socks in the drawer

Worked out on the board in the lab.

?

1. What is the likelihood $L(\pi)$?
2. Find $\hat{\pi}_{\text{ML}}$.
3. Find $\hat{\pi}_{\text{MAP}}$ with a prior $\beta(\pi \mid a, b)$.
4. Does it matter if T is an **ordered** sequence?

More problems like this: exercise book on CourseWare.

Socks

A drawer with **red** and **green** socks. We pull out a sock, note its colour and **put it back**. After N draws, R socks were red.

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{\text{red}, \text{green}\}$$

$$p(\text{red} \mid \pi) = \pi, \qquad p(\text{green} \mid \pi) = 1 - \pi$$

π: the ratio of red socks in the drawer

Worked out on the board in the lab.

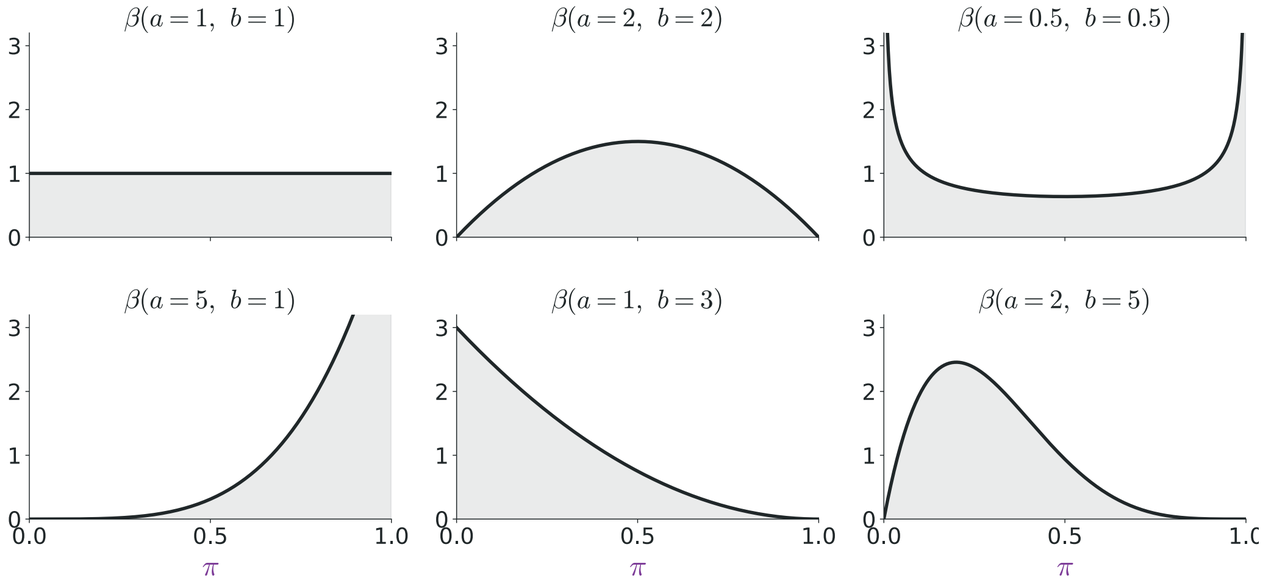
$$\hat{\pi}_{\text{ML}} = \frac{R}{N}, \qquad \hat{\pi}_{\text{MAP}} = \frac{R + a - 1}{N + a + b - 2}$$

?

1. What is the likelihood $L(\pi)$?
2. Find $\hat{\pi}_{\text{ML}}$.
3. Find $\hat{\pi}_{\text{MAP}}$ with a prior $\beta(\pi \mid a, b)$.
4. Does it matter if T is an **ordered** sequence?

More problems like this: exercise book on CourseWare.

The Beta prior: virtual observations

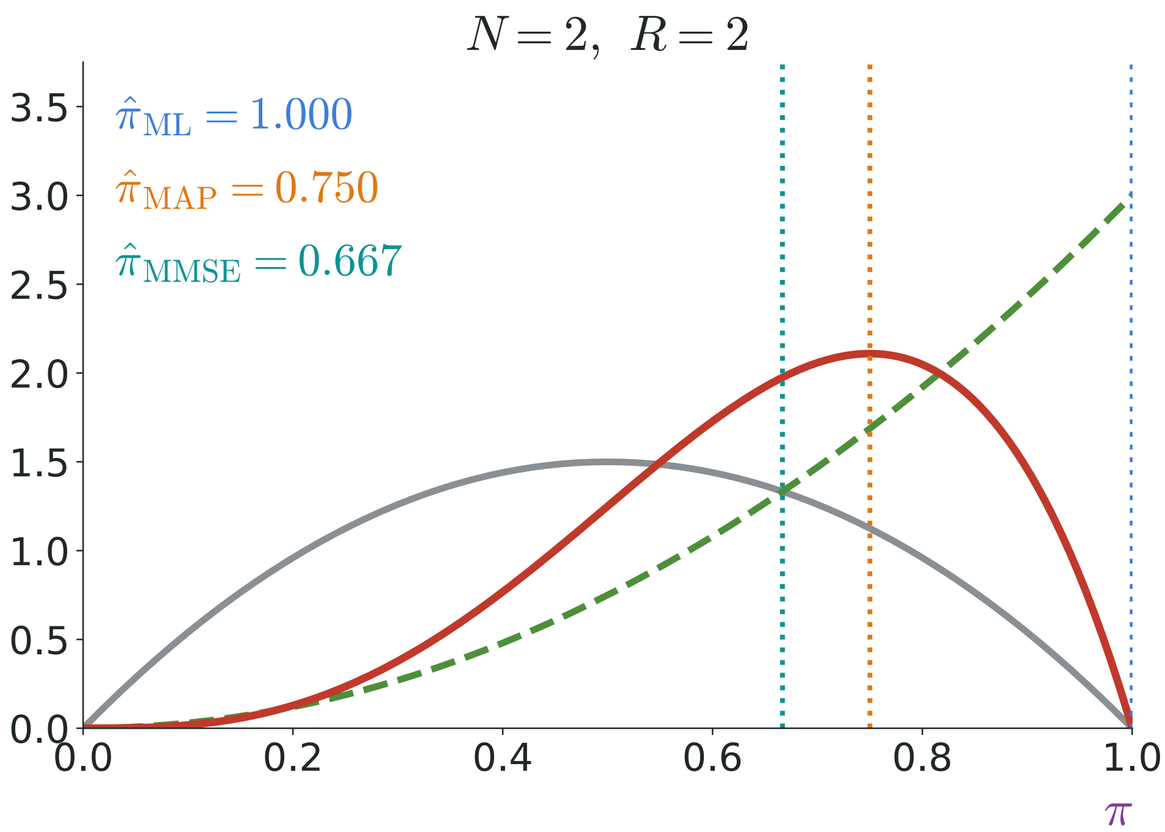


$$\hat{\pi}_{\text{MAP}} = \frac{R + a - 1}{N + a + b - 2}$$

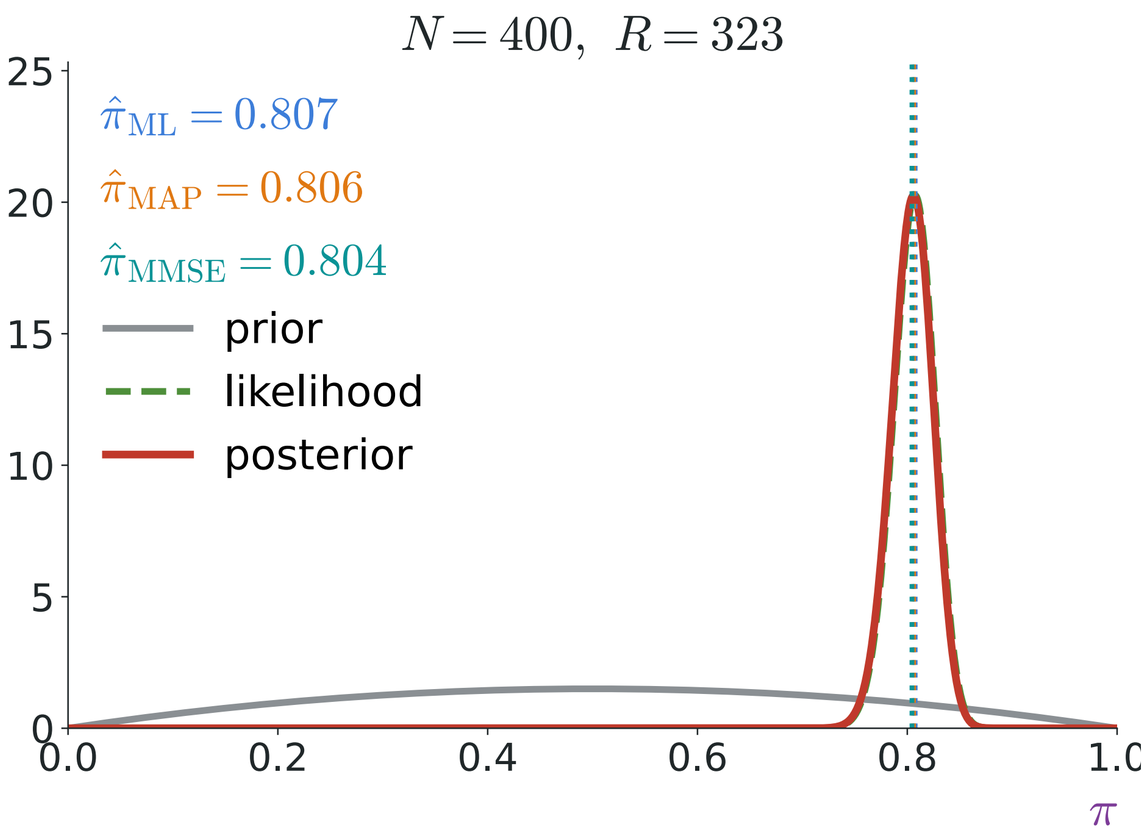
> ! As if we had seen $a - 1$ extra **red** and $b - 1$ extra **green** socks.

a, b are **hyperparameters**: parameters of the prior.

Socks in numbers



Two red socks: ML says **all** socks are red.



400 socks: the prior **does not matter** any more.

Socks: try it

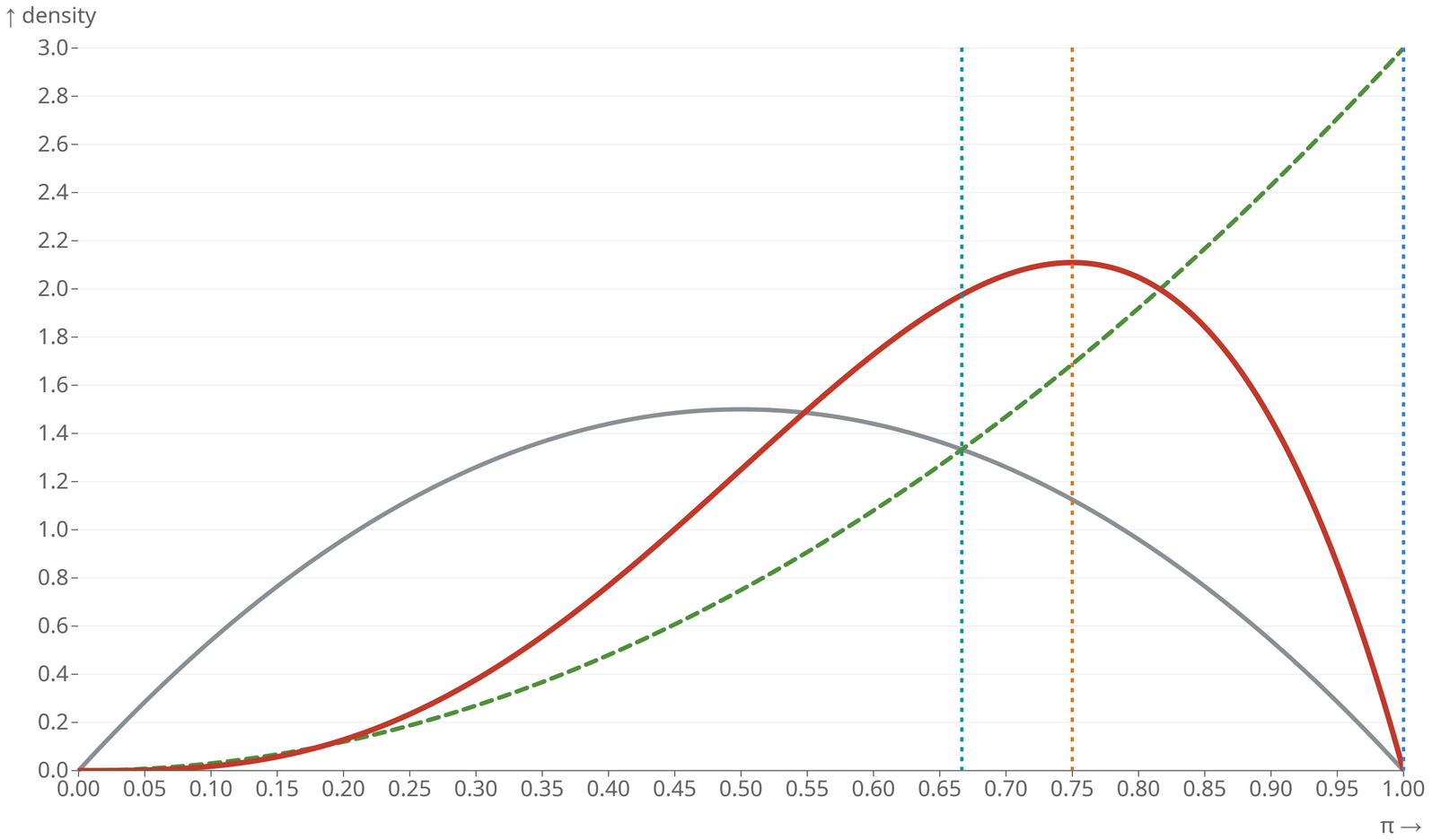
N socks	2
R red	2
prior a	2
prior b	2

$\pi_{ML} = 1.000$

$\pi_{MAP} = 0.750$

$\pi_{MMSE} = 0.667$

prior, likelihood, posterior



Categorical distribution

A die with K sides

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{1, \ldots, K\}$$

$$p(x = k \mid p_1, \ldots, p_K) = p_k$$

$$\sum_{k=1}^{K} p_k = 1, \qquad N_k = \text{count of } k$$

Worked out on the board in the lab.

? 1. Write the likelihood $L(p_1, \ldots, p_K)$.
2. Find $\hat{p}_k^{\mathrm{ML}}$. Do not forget the **constraint**!

Hint: use a **Lagrange multiplier** for the constraint $\sum_k p_k = 1$.

More problems like this: exercise book on CourseWare.

A biased die, an election, words in a text.

A die with K sides

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{1, \ldots, K\}$$

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$$\sum_{k=1}^{K} p_k = 1, \qquad N_k = \text{count of } k$$

Worked out on the board in the lab.

$$\hat{p}_k^{\mathrm{ML}} = \frac{N_k}{N} \text{ (the relative frequency)}$$

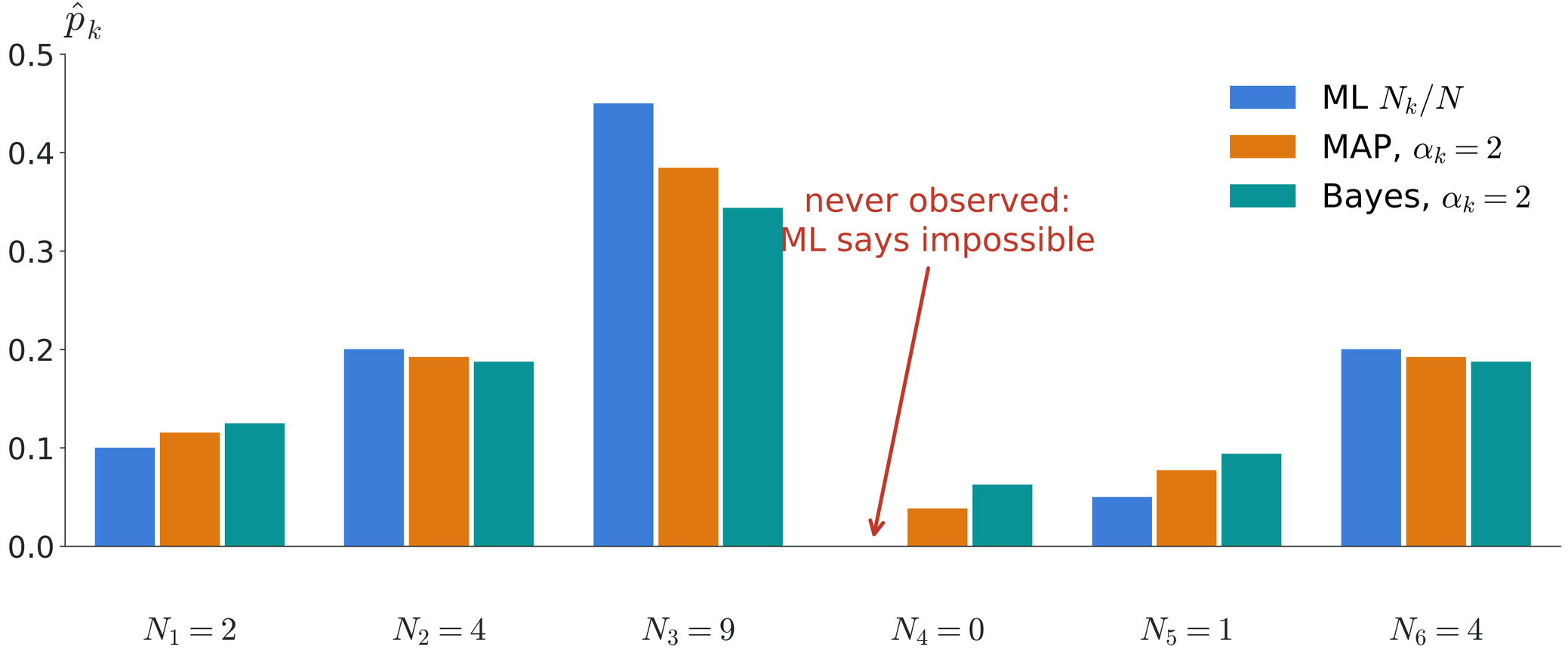
? 1. Write the likelihood $L(p_1, \ldots, p_K)$.
2. Find $\hat{p}_k^{\mathrm{ML}}$. Do not forget the **constraint**!

Hint: use a **Lagrange multiplier** for the constraint $\sum_k p_k = 1$.

More problems like this: exercise book on CourseWare.

A biased die, an election, words in a text.

A category never observed



ML gives an unseen category probability **0**. The prior gives it a small chance.

Practice

Exponential distribution

Waiting times $T = \{t_1, \ldots, t_N\}$, exponential distribution:

$$p(t \mid \lambda) = \frac{1}{\lambda} e^{-t/\lambda}, \qquad t \geq 0$$

λ: the mean waiting time

Try it yourself, then check your result:

? 1. Find $\hat{\lambda}_{\mathrm{ML}}$.
2. Substitute $\theta = 1/\lambda$ and find $\hat{\theta}_{\mathrm{ML}}$. What is the relation?

More problems like this: exercise book on CourseWare.

Exponential distribution

Waiting times $T = \{t_1, \ldots, t_N\}$, exponential distribution:

$$p(t \mid \lambda) = \frac{1}{\lambda} e^{-t/\lambda}, \qquad t \geq 0$$

λ: the mean waiting time

Try it yourself, then check your result:

$$\hat{\lambda}_{\mathrm{ML}} = \frac{1}{N} \sum_i t_i, \qquad \hat{\theta}_{\mathrm{ML}} = \frac{1}{\hat{\lambda}_{\mathrm{ML}}}$$

1. Find $\hat{\lambda}_{\mathrm{ML}}$.
2. Substitute $\theta = 1/\lambda$ and find $\hat{\theta}_{\mathrm{ML}}$. What is the relation?

More problems like this: exercise book on CourseWare.

The assignment

Assignment: MLE, MAP, Bayes

Part 1: normal

- ML of μ, σ^2
- MAP: **normal-inverse-gamma** prior
- Bayes: posterior and **predictive** density

Part 2: categorical

- ML: relative frequencies
- MAP: **Dirichlet** prior
- Bayes: predictive $\frac{N_k+\alpha_k}{N+\sum_j \alpha_j}$

Part 3: classifier

Letters **A and C**, training sets of 20, 200 and 2000 samples.

Compare ML, MAP and Bayes on the test set.

Take-away for the lab

- ☑ Known form $p(x \mid \theta)$: estimate the **parameters** from i.i.d. samples
- ☑ **ML**: $\arg\max_\theta p(T \mid \theta)$, simple, the best with a lot of data
- ☑ **MAP**: $\arg\max_\theta p(T \mid \theta)\, p(\theta)$, the prior as **virtual observations**
- ☑ **Bayes**: the whole posterior, prediction **averages** all models
- ☑ **MMSE**: the mean of the posterior
- ☑ With few data the choice **matters**: socks, unseen categories, German tanks


"That's all Folks!"